

♡

Regression logistica → Separa 2 classes
One vs Rest ↑

Trajectory 1

$$x = [0 \text{ a } 10]$$

$$x'(t) = 1$$

$$C(x) = 2 \sin(x) e^{0.3x}$$

$$C' = 2 \sin(x) e^{\frac{3}{10}x}$$

$$C' = 2 e^{\frac{3}{10}x} \sin(x)$$

$$C' = \underbrace{2}_{u'} \cdot \underbrace{e^{\frac{3}{10}x}}_v$$

$$h'(x) = u'(x) \cdot v(x) + v(x) \cdot v'(x)$$

$$= u' = 2 \sin(x) \quad v = e^{0.3x / \frac{3}{10}x}$$

$$= v'(x) = e^{0.3x} \cdot 0.3$$

$$f'(x) = \underbrace{(2 \cos(x))}_{u'} \cdot \underbrace{e^{0.3x}}_v + \underbrace{(2 \sin(x))}_{u'} \cdot \underbrace{0.3 e^{0.3x}}_{v'}$$

$$0.6 \sin(x) e^{0.3x} + 2 \cos(x) e^{0.3x}$$

$$2 e^{0.3x} \left(\frac{0.6 \sin(x)}{2} + \cos(x) \right)$$

$$y'(t) \quad \cancel{C'(x)} = 2 e^{0.3x} (\cos(x) + 0.3 \sin(x)) \quad \checkmark$$

$$u(t) = \sqrt{1 + (2 e^{0.3x} (\cos(x) + 0.3 \sin(x)))^2}$$

1(0.01)

$$\theta(t) = \tan^{-1}\left(\frac{\dot{y}}{\dot{x}}\right) = \tan^{-1}\left[\frac{2e^{0.3x}(\cos(x) + 0.3\sin(x))}{1}\right]$$

$$\omega(t) = \frac{d}{dt} \tan^{-1}\left(2e^{0.3x}(\cos(x) + 0.3\sin(x))\right)$$

$$\hookrightarrow \frac{d}{dt} \tan^{-1}(u) = \frac{1}{1+u^2} \frac{du}{dt}$$

$$\frac{1}{1 + \left(2e^{0.3x}(\cos(x) + 0.3\sin(x))\right)^2} \cdot \frac{du}{dt}$$

$$\frac{d}{dt} (2e^{0.3x}(\cos(x) + 0.3\sin(x)))$$

$$\bullet \quad v = 2e^{0.3x} \cdot 0.3 = 0.6e^{0.3x} \quad v' = (-\sin(x) + 0.3\cos(x))$$

$$f' = [0.6e^{0.3x}(\cos(x) + 0.3\sin(x))] + [2e^{0.3x}(-\sin(x) + 0.3\cos(x))]$$

$$\frac{du}{dt} = \frac{e^{0.3x}(1.2\cos(x) - 1.8\sin(x))}{1 + (2e^{0.3x}(\cos(x) + 0.3\sin(x)))^2}$$

$$\frac{du}{dt} = \frac{e^{0.3x}(1.2\cos(x) - 1.8\sin(x))}{1 + (2e^{0.3x}(\cos(x) + 0.3\sin(x)))^2} \quad (2)$$

$$1) u(t) = \sqrt{1 + (x)^2}$$

$$2) v(t) = \sqrt{1 + (-2x + 10)^2}$$

$$3) u(t) = \sqrt{1 + (2e^{x-7}(\cos(x-7) - \sin(x-7)))^2}$$

$$2) x=1$$

$$(2x = -(x-5)^2 + 8 = -2(x-5))$$

$$= -2x + 10$$

$$3) x=1$$

$$f_3(x) = 2e^{(x-7)} \cos(x-7)$$

$$u = 2e^{x-7}$$

$$v = \cos(x-7)$$

$$u'v + uv' = (2e^{x-7}) \cos(x-7) + (2e^{x-7})(-\sin(x-7))$$

$$= 2e^{x-7} (\cos(x-7) - \sin(x-7))$$

$$1) \theta(t) = \tan^{-1} \left(\frac{x(t)}{1} \right)$$

$$w(t) = \frac{d}{dt} \left(\tan^{-1} \left(\frac{x(t)}{1} \right) \right)$$

$$= \frac{1}{1 + \left(\frac{x}{1} \right)^2} \cdot \frac{d \left(\frac{x}{1} \right)}{dt} = \frac{1}{1 + \left(\frac{x}{1} \right)^2} \cdot 1 =$$

$$= \frac{1}{1 + x^2}$$

$$2) \phi(t) = \tan^{-1} \left(\frac{-2x+10}{1} \right)$$

$$w(t) = \frac{d}{dt} \left(\tan^{-1} \left(\frac{-2x+10}{1} \right) \right) \cdot \frac{1}{1 + (-2x+10)^2}$$

$$w(t) =$$

$$\frac{-2}{4x^2 - 40x + 101}$$

$$3) \frac{1}{1 + (2e^{x-7} (\cos(x-7) - \sin(x-7)))^2} \cdot -4e^{x-7} \sin(x-7)$$

← has to be on
photostat

$$w(t) = \frac{-4e^{x-7} \sin(x-7)}{1 + 4e^{2x-14} (1 - \sin(2x-14))}$$